

Assignment On

‘ADVANCED DATA STRUCTURES AND ALGORITHMS’

(Assignment-2)

Submitted by:
Madhu boini
2503B09904(MCA).

Submitted to:
Dr. Rahul Mishra.

Question: Convert The Patient List to A Circular Linked List for a Round-Robin Check-Up System. Implement Insertion and Deletion.

Code :

```
class Patient:
    def __init__(self, name, age, patient_id):
        self.name = name
        self.age = age
        self.id = patient_id
        self.next = None

class CircularPatientList:
    def __init__(self):
        self.head = None

    # Insert patient at end (circular)
    def insert_patient(self, name, age, patient_id):
        new_patient = Patient(name, age, patient_id)

        if self.head is None:
            self.head = new_patient
            new_patient.next = self.head
        else:
            temp = self.head
            while temp.next != self.head:
                temp = temp.next
            temp.next = new_patient
            new_patient.next = self.head

        print("Patient inserted:", name)

    # Delete patient by ID
    def delete_patient(self, patient_id):
        if self.head is None:
            print("List is empty.")
```

```

        return

temp = self.head
prev = None

# Case 1: Only one node
if temp.id == patient_id and temp.next == self.head:
    self.head = None
    print("Only patient deleted. List empty now.")
    return

# Case 2: Head deletion with more nodes
if temp.id == patient_id:
    # Find last node
    last = self.head
    while last.next != self.head:
        last = last.next

    last.next = self.head.next
    self.head = self.head.next
    print("Head patient deleted with ID:", patient_id)
    return

# Case 3: Delete non-head node
while temp.next != self.head:
    prev = temp
    temp = temp.next
    if temp.id == patient_id:
        prev.next = temp.next
        print("Patient deleted with ID:", patient_id)
        return

print("Patient not found with ID:", patient_id)

# Display circular list
def display(self):
    if self.head is None:
        print("No patients in the list.")
        return

    print("\n--- Round Robin Patient List ---")
    temp = self.head
    while True:
        print(f"Name: {temp.name}, Age: {temp.age}, ID: {temp.id}")
        temp = temp.next
        if temp == self.head:
            break

# -----
# Example Usage
# -----

```

```
cplist = CircularPatientList()

cplist.insert_patient("Ravi", 30, 101)
cplist.insert_patient("Sita", 25, 102)
cplist.insert_patient("Kiran", 40, 103)

cplist.display()

cplist.delete_patient(102)

cplist.display()
```

Output:

```
PS C:\Users\Madhu\OneDrive\Pictures\Desktop\ADSA\Assignment-2> & 'C:\Users\Madhu\AppData\Local\Programs\Python\Python314\python.exe' 'c:\Users\Madhu\.vscode\extensions\ms-python.debugpy-2025.16.0-win32-x64\bundled\libs\debugpy\launcher' '58463' '--' 'C:\Users\Madhu\OneDrive\Pictures\Desktop\ADSA\Assignment-2\task.py'
Patient inserted: Ravi
Patient inserted: Sita
Patient inserted: Kiran

--- Round Robin Patient List ---
Name: Ravi, Age: 30, ID: 101
Name: Sita, Age: 25, ID: 102
Name: Kiran, Age: 40, ID: 103
Patient deleted with ID: 102

--- Round Robin Patient List ---
Name: Ravi, Age: 30, ID: 101
Name: Kiran, Age: 40, ID: 103
PS C:\Users\Madhu\OneDrive\Pictures\Desktop\ADSA\Assignment-2>
```